

EDUCATION

Brown University

Providence, RI

Sc. B Computer Science, Sc. B Mathematics

2021 - 2025

- GPA: 3.91/4.00, Magna Cum Laude
- Honors Thesis: Distributed Quantum Signal Processing. (External Advisor: Yuan Liu, NC Stat; Reader: Brenda Rubenstein, Brown, Department Advisor: James Tompkin, Brown)

RESEARCH INTERESTS

Quantum algorithms and complexity theory, mathematical structures in quantum computing, quantum error correction

RESEARCH
EXPERIENCE**Postbaccalaureate Researcher (Remote), North Carolina State University, Remote**
Jul '25 - Present

- Part-time continuation of thesis research after graduation, advised by Prof. Yuan Liu and in collaboration with researchers from the Yale Quantum Institute, QuEra, and MIT.
- Focused on distributed quantum algorithms.

SULI Program, Oak Ridge National Lab, Oak Ridge, TN

May '23 - Jul. '23

- Developed methods for solving constrained optimization problems using the Quantum Approximate Optimization Algorithm (QAOA) advised by Dr. Phillip Lotshaw.
- Mathematically proved correctness and conducted simulations to demonstrate strong performance.

Undergraduate Researcher, Brown Visual Computing Lab, Providence, RI May '22 - May. '23

- Advised by Prof. James Tompkin
- **Decoupled Style Descriptors Project:** Built an interactive application for synthesizing handwriting via a recurrent neural network.
- **Augmented Reality Label Visualization Project:** Implemented and optimized algorithms for adapting augmented reality content in real-time to dynamic environments using high-performance Unity shaders.

PAPERS

PREPRINTS

1. **Brayden Goldstein-Gelb**, Kun Liu, John Martyn, Yongshan Ding, Yuan Liu. Distributed Quantum Signal Processing. *In Preparation*.
2. **Brayden Goldstein-Gelb**, Phillip C. Lotshaw. Convergence guarantee for linearly-constrained combinatorial optimization with a quantum alternating operator ansatz. *Under review*, ACM TQC (submitted Dec 2024); preprint available at [arXiv:2409.18829](https://arxiv.org/abs/2409.18829) (Sept 2024).

POSTERS

1. Ashley Kwon, Yuanbo Li, Eva Schiller, **Brayden Goldstein-Gelb**, James Tompkin. Environment Adaptive AR Label Visualization. Brown University Undergraduate Research Symposium, April 2023. Poster presentation.

COURSEWORK

- **Computer Science:** An Algorithmist's Toolkit (grad.), Design and Analysis of Algorithms, Theory of Computation, Algorithmic Machine Learning, Cryptography, Systems, Formal Methods, Computer Vision
- **Mathematics:** Four semesters of Algebra (two undergrad, two grad), Real Analysis, Complex Analysis, Graph Theory, Number Theory, Calculus on Manifolds, Linear Algebra, Mathematical Quantum Mechanics, Information Theory, Optimization and Stochastic Calculus, Statistical Inference
- **Physics/Engineering:** Quantum Mechanics, Quantum Information, Dynamics and Vibrations, Mechanics

PROFESSIONAL EXPERIENCE	Software Engineer (Incoming), NewGrid Inc. , Somerville, MA	Aug '25 – Present
	Returning full-time after internship	
	Software Engineering Intern, NewGrid Inc. , Somerville, MA	May '24 – Aug '24
	- Built real-time grid outage monitoring and visualization tools using Vue.js and D3. - Automated data ingestion pipelines to integrate outage data with internal tools.	
TEACHING EXPERIENCE	Grader , Linear Algebra with Theory	Spring '25
	Graded problem sets for MATH 0540	
	Teaching Assistant , Theory of Computation	Fall '24
	Created problem sets and example solutions, assisted in grading, held weekly office hours	
	Online Tutor , Wyzant Inc.	May '21 – Aug '23
	- Over 350 hours of tutoring experience, working with 50+ students ranging from middle schoolers to executives. - Maintained a five-star rating, helping students build confidence and mastery in Python, Java, Calculus, and Algebra.	
PROJECTS	Hackathon Projects	
	<i>MIT iQuHack, Brown Quantum Hackathon, Hack@Brown</i>	Fall '24
	- Won 2nd Place at Brown Quantum Hackathon - Completed QuEra challenge at MIT's iQuHack quantum hackathon - Created a space-themed game for Hack@Brown	
	Quantum Mechanics Proofs Library	
	<i>Final Project, Formal Proof and Verification</i>	Fall '22
	- Created a library in the Lean Theorem Prover to model quantum particles, measurements, and multi-particle states. - Demonstrated the library's capabilities by constructing a formal proof of the no-cloning theorem.	
	Re-Font Chrome Extension	
	<i>JavaScript, Chrome APIs</i>	Spring '22
	- Published a Chrome extension that allows users to customize website appearance and improve accessibility, with over 9,300 total installs and 750 current active users. - Continuously improved the user interface based on feedback, enhancing user experience and accessibility.	
	Additional personal and academic projects available on my website	
AWARDS	• Sigma Xi , Scientific Research Honor Society	2025
	• Audience Favorite Award , Brown Undergraduate Research Symposium	2023
SKILLS	Programming Languages: Python, C/C++, Java, HTML/CSS, JavaScript, Julia, Pyret	
	Technologies: Qiskit, Git, NumPy, Sklearn, Pandas, TensorFlow, PyTorch, Flask, PostgreSQL, Vue, D3, Node.js	
	Updated: July 2025	